

FUNCTIONAL REQUIREMENT DOCUMENT

Insight Insitute

Phase 1 — Core Modules

Document Version	1.0
Status	Final Draft
Date	May 2026
Total Modules	13 (9 Core + Chat + Notifications + Audit + Reporting)
Platforms	Web (Admin Panel) + Mobile App (iOS & Android)
User Roles	15 Roles

1. Introduction

1.1 Purpose

This Functional Requirement Document (FRD) defines the complete functional scope, business rules, user roles, and system behaviour for Phase 1 of the Insight Institute. It serves as the authoritative reference for design, development, quality assurance, and stakeholder sign-off. Every feature described herein must be implemented, tested, and accepted before Phase 1 is considered complete.

1.2 Project Overview

The ERP is a multi-platform (web + mobile) system designed to digitise and automate the end-to-end operations of an educational institution — from first student inquiry through to payroll processing. The system will serve 15 distinct user roles across multiple branches, each with precisely scoped access and capabilities.

1.3 Scope of Phase 1

Phase 1 encompasses 13 modules delivered on two platforms:

#	Module	Platform
1	CRM & Pre-Admission	Web + Mobile
2	Student Management	Web + Mobile
3	Timetable & Scheduling	Web + Mobile
4	Attendance Management	Web + Mobile
5	Fees Management	Web + Mobile
6	Exam Management	Web + Mobile
7	Exam Supervision	Web + Mobile
8	Faculty & Payroll	Web + Mobile
9	Leave Management	Web + Mobile
10	In-App Chat	Mobile + Web
11	Notification System	Mobile + Web
12	Audit Logs	Web (Admin)
13	Reporting & Dashboard	Web + Mobile

2. System Architecture Overview

2.1 Platform Components

Component	Description
Web Admin Panel	Browser-based interface for admin, management, finance, and HR roles. Full feature access with rich data entry forms, dashboards, and reports.
Mobile Application (Student)	iOS & Android app — timetable, attendance QR, exams, fee status, chat, notifications, digital ID.
Mobile Application (Parent)	iOS & Android app — child attendance, fee payments, exam results, chat with admin, ticket submission.
Mobile Application (Faculty)	iOS & Android app — timetable, exam creation, paper upload, attendance QR, session reporting, leave, payroll slips.
Backend API	RESTful API server serving all platforms. Single source of truth for business logic.
Push Notification Service	Firestore Cloud Messaging (Android) + APNs (iOS) for real-time push delivery.
Storage	Cloud object storage for documents, receipts, exam papers, payment screenshots, chat file attachments.

2.2 Authentication & Security

- Role-Based Access Control (RBAC) — each role has a defined permission matrix (see Section 4).
- Multi-branch data isolation — Branch Manager and below can only access their own branch data.
- All API endpoints require JWT authentication with role-claim validation.
- All file uploads (payment screenshots, exam papers) are stored securely with access control.
- Audit trail is maintained for sensitive operations: fee approval, record modification, leave decisions.

3. User Roles & Permission Matrix

The system implements 15 distinct user roles governed by Role-Based Access Control (RBAC). Each role is described below with full access grants and explicit restrictions.

3.1 Super Admin

Highest authority across all branches. No restrictions whatsoever.

Access Granted:

- All modules and system configuration
- User management and permission settings across all branches
- Full financial data: fees, payments, refunds, payroll
- All reports and analytics
- Audit logs viewer

Restrictions:

- None — unrestricted access

3.2 Branch Manager

Controls and oversees a single assigned branch.

Access Granted:

- CRM, Students, Parents, Faculty management
- Attendance and timetable oversight
- Fees overview (read + approve cash entries raised by Admin Senior Executive)
- Exam module access
- Leave approvals for branch staff
- Branch-level reports
- Refund approval (Accountant must have Branch Manager approval for any refund)

Restrictions:

- Cannot access data of other branches
- Cannot modify system-wide configuration

3.3 Admin Senior Executive

Operational head / admin coordinator for a branch.

Access Granted:

- Student admission workflow and batch allocation
- Attendance oversight
- Timetable management — create, update, and publish; notify all affected users on change
- Fee follow-up and cash fee entry (submitted for Branch Manager approval)

- Exam coordination
- Paper rechecking workflow management
- Branch-level reports
- Approve / reject leave applications

Restrictions:

- No payroll access
- Cannot delete records — edits only

3.4 Admin Executive

Day-to-day data entry operator.

Access Granted:

- Student admission form entry
- Attendance data entry and QR operations
- Basic timetable view (read-only)

Restrictions:

- No delete permissions on any record
- No payroll access
- No reports module

3.5 Front Desk Executive

Reception and inquiry intake only.

Access Granted:

- Inquiry form fill and view (CRM entry-point only)
- Assign lead/student to Counsellor or Sales Executive

Restrictions:

- All other modules are fully restricted

3.6 Counsellor

Manages assigned leads through the sales/admission pipeline.

Access Granted:

- View only their assigned leads
- Change lead status: Confirm, Add Notes, Payment Process, Payment Received, Payment Approve

Restrictions:

- Cannot see unassigned or other Counsellor's leads
- No student, academic, or financial module access

3.7 Telecaller

Outreach and lead contact role.

Access Granted:

- Assigned leads only — call logging and status updates
- Basic CRM inquiry view for assigned records

Restrictions:

- No access beyond assigned lead records

3.8 Sales Senior Executive

Full CRM pipeline authority.

Access Granted:

- Complete CRM access — all leads, pipelines, and conversion reports
- Can reassign leads and view team performance

Restrictions:

- No access to academic, financial, or operational modules

3.9 Sales Executive

Lead assignment only.

Access Granted:

- View and manage assigned inquiries only

Restrictions:

- No CRM-wide access
- No academic or financial module access

3.10 Student

Enrolled student using the mobile app.

Access Granted:

- Personal dashboard
- Own attendance records
- Own timetable
- Online exam taking
- Own exam results

- Own fee status
- Push notifications and in-app notification center
- Chat with faculty and admin
- QR-based digital ID / entry pass

Restrictions:

- Cannot view other students' data
- No management or admin features

3.11 Parent

Guardian linked to one or more enrolled students.

Access Granted:

- Child's attendance
- Fee payments and receipt downloads
- Child's exam results
- Child's timetable
- Push notifications
- Chat with Super Admin, Branch Manager, Admin Senior Executive
- Raise support tickets (visible to Super Admin, Branch Manager, Admin Senior Executive)

Restrictions:

- Only linked student's data visible
- No management features

3.12 Faculty

Teaching staff member.

Access Granted:

- Own timetable
- Exam paper upload and online exam creation
- Attend via QR check-in/check-out
- Session reporting (subject, chapter, completion status)
- Leave application submission
- Group chat channels only (no one-to-one direct chat)
- Own payroll slips

Restrictions:

- No fee or admissions module access
- No direct/private chat (group channels only)

3.13 Exam Supervisor

On-ground exam operations role.

Access Granted:

- Exam schedules and seating arrangements
- Student attendance during exams
- QR validation at exam hall entry
- Malpractice report submission
- Own payroll slips
- Exam supervision dashboard

Restrictions:

- No academic module
- No finance module

3.14 Paper Checker

Evaluation-only role for offline exam papers.

Access Granted:

- Assigned papers (automatic assignment by system)
- Marks entry and remarks
- Wrong-answer query raised with screenshot (visible to Branch Manager and Admin Senior Executive)
- Paper rechecking workflow (triggered by student, routed through Admin Senior Executive)
- Result submission
- Own payroll slips (auto-computed for assigned papers)

Restrictions:

- No student profile access beyond exam-related info
- No fee, attendance, or payroll configuration

3.15 Accountant

Finance and billing role.

Access Granted:

- Fees and billing management
- Payment verification and approval
- Refund processing (requires Branch Manager approval; refund transaction ID or cheque number mandatory)
- Payroll computation and disbursement
- Financial reporting and reconciliation
- Receipt generation and dispatch to Student and Parent

Restrictions:

- No academic operations

- No exam management module

4. Functional Requirements — Module Detail

4.1 Module 1 — CRM & Pre-Admission

Manages the full lifecycle of prospective student inquiries from initial capture to admission conversion or disqualification.

4.1.1 Inquiry Capture

- Front Desk Executive and authorised roles can create a new inquiry record from the web admin panel or mobile app.
- Inquiry form fields: Student name, guardian name, contact number, email, course interested in, source (walk-in / referral / online / telecall), date of inquiry, remarks.
- System auto-assigns a unique Inquiry ID and timestamps the entry.
- Front Desk Executive may assign the lead to a Counsellor or Sales Executive at creation.

4.1.2 Lead Pipeline

All inquiries progress through a defined pipeline:

- New — inquiry just created, not yet contacted.
- Contacted — at least one interaction logged.
- Interested — expressed intent to enrol.
- Converted — admission process initiated (one-click conversion).
- Lost — not proceeding; reason must be recorded.

4.1.3 Interaction History

- Each lead maintains a chronological log of all interactions: call notes, visit notes, follow-up tasks, status changes.
- Counsellors and Telecallers can add notes and update status only for leads assigned to them.
- Sales Senior Executive can view and reassign any lead across the team.

4.1.4 One-Click Admission Conversion

- When a lead is moved to Converted, the system auto-populates the Student Admission form (Module 2) with available data from the inquiry record.
- Admin Senior Executive or Branch Manager finalises the admission after reviewing the pre-filled form.

4.1.5 Role-Specific CRM Behaviour

Role	CRM Behaviour
Front Desk Executive	Create inquiry, view all inquiries, assign to Counsellor / Sales Executive
Counsellor	See only assigned leads; change status, add notes, update payment status
Telecaller	See and update only assigned leads
Sales Executive	See and manage assigned inquiries only

Role	CRM Behaviour
Sales Senior Executive	Full CRM access — all leads, reassignment, team performance view
Branch Manager	Full CRM view for own branch
Super Admin	Full access across all branches

4.2 Module 2 — Student Management

Covers the complete lifecycle of a student record from admission through graduation, including linking to parents, inventory, and digital identity.

4.2.1 Online Admission Form

- Multi-step form: Personal details, academic background, guardian/parent info, course/batch selection, document uploads (photo ID, previous marksheets), emergency contact.
- System validates mandatory fields and enforces document upload before finalising.
- On submission, an Admission Number is auto-generated and the student status is set to Active.

4.2.2 Student Master Profile

- Centralised profile containing: personal info, contact details, enrolled course/batch, fee payment history, attendance summary, exam results, leave records, digital ID, linked parents.
- Profile photo is mandatory and used on the digital ID card.
- Admin can flag a student as Inactive, Transferred, or Alumni.

4.2.3 Parent Linking

- One student may be linked to one or more parents/guardians.
- Each parent gets an independent mobile app login with access restricted to their linked child/children only.
- Parent portal is auto-activated on admission; credentials are dispatched via in-app notification.

4.2.4 Batch Allocation

- Admin Senior Executive allocates a student to a batch at admission time.
- Batch changes are logged with the reason, acting user, and timestamp.

4.2.5 Inventory Tracking — Uniform, Books, QR

- Admin can issue uniform items (size, quantity, date issued) and books (title, ISBN, quantity) per student.
- Each issue is logged for inventory reconciliation.
- QR-code-enabled physical ID card is associated with the student's digital profile.

4.2.6 Digital ID Card

- System generates a digital ID card containing: student name, admission number, course, batch, photo, QR code.
- QR code encodes the student's unique system ID and is used for attendance check-in/check-out (Module 4).
- Students can display their digital ID from the mobile app; app-generated QR is accepted at check-in points.

4.3 Module 3 — Timetable & Scheduling

4.3.1 Academic Calendar

- Admin Senior Executive defines the academic year, terms, exam periods, public holidays, and institution holidays.
- Calendar is published to all users and reflected in timetable scheduling.

4.3.2 Batch-Wise Timetable

- Each batch has an independent weekly timetable specifying: day, period slot, subject, faculty, classroom.
- Admin Senior Executive creates and manages timetables.

4.3.3 Drag-and-Drop Scheduler

- Web admin panel provides a visual drag-and-drop grid for scheduling periods.
- Faculty and classroom are selected from dropdowns filtered by availability.

4.3.4 Conflict Detection

- System automatically detects and alerts on: Faculty double-booking (same faculty in two batches at same slot), Classroom double-booking, Subject overload (exceeds weekly hour cap for a subject).
- Conflicting slots are highlighted; the admin must resolve conflicts before publishing.

4.3.5 Revised Timetable Notifications

- When a published timetable is modified, the system auto-sends push notifications to all affected Students and Faculty.
- Admin Senior Executive creates the updated timetable; notification dispatched automatically on publish.

4.3.6 Classroom Utilisation Reports

- Reports show: occupancy rate per classroom per week, free slots, over-utilised rooms.

4.4 Module 4 — Attendance Management

4.4.1 QR Code Check-In / Check-Out

- Students scan their digital QR ID at the institution's QR reader (or a mobile scan point) upon entry and exit.
- Each scan logs: student ID, scan type (IN / OUT), timestamp, device ID.
- In-app QR can be displayed from the student mobile app for scanning.

4.4.2 Delay Alerts

Alert Trigger	Recipient & Action
Student has not checked in 15 minutes after session start	Push notification to Parent — '15-minute delay alert'
Student has not checked in 30 minutes after session start	Push notification to Parent AND Admin — '30-minute escalation alert'
Missing scan — no OUT scan recorded by end of day	Push notification to Parent — 'Missing check-out scan' alert
Missing scan — no IN scan recorded for scheduled session	Push notification to Parent — 'Absent / Missing check-in' alert

4.4.3 Violation Management

- Admin can log attendance violations (repeated delay, unauthorised absence) against a student's record.
- Violations are visible in the student profile and can be included in reports.

4.5 Module 5 — Fees Management

4.5.1 Fee Structure Setup

- Accountant / Admin Senior Executive configures fee structures per course: admission fee, tuition fee, exam fee, miscellaneous charges.
- Installment schedules are defined: due date, amount, late penalty rules.

4.5.2 Online Payment Upload Workflow

- Student or Parent uploads a payment screenshot from the mobile app.
- Accountant receives a notification, verifies the screenshot against the actual bank record, and approves or rejects.
- On approval, the payment is posted to the student's account and a digital receipt is generated and dispatched (push notification + in-app).

4.5.3 Cash Payment Workflow

- Admin Senior Executive creates a cash payment entry against a student's account.
- Entry is submitted to Branch Manager for approval.
- On Branch Manager approval, the payment is posted and reflected in the student's total fees paid.
- Digital receipt is generated and sent to Student and Parent.

4.5.4 Automated Reminders

- System sends automated fee due reminders to Student and Parent 7 days, 3 days, and 1 day before each installment due date.
- Overdue fee alerts are sent daily until payment is received.

4.5.5 Refund Workflow

- Accountant initiates a refund request.
- Branch Manager must approve the refund.
- A refund transaction ID (bank transfer) or cheque number is mandatory before the refund can be posted.

4.6 Module 6 — Exam Management

4.6.1 Online Exam — Mobile

Feature	Requirement
Timer	Countdown timer visible throughout; auto-submits when time expires.
Single Attempt	Only one submission allowed per student per exam. Re-entry after submission is blocked.
Autosave	Answers are saved to the server every 30 seconds and on each question navigation to prevent data loss.
Geo-Fence	Exam can be geo-restricted to the institution premises (configurable radius). Location check at exam start and periodically during exam.
Screen Lock	Mobile app prevents switching to other apps during an active exam. Screen-lock violation is logged.
Split-Screen Detection	Detected and logged as a potential malpractice event. Configurable to auto-submit or flag.
Instant Results	On submission, results are computed and displayed immediately (for objective exams). Faculty can configure delayed results.

4.6.2 Offline Exam Workflow

- Admin Senior Executive / Faculty creates an offline exam event (date, time, batch, subject, duration).
- Paper Checkers are assigned to the exam by Admin Senior Executive.
- Checker receives a secure link / in-app notification to access assigned answer sheets.
- Admin tracks submission status of all assigned papers on a tracking dashboard.
- Paper rechecking: Student raises a recheck request → routed to Admin Senior Executive → assigned to a Checker → recheck result submitted and published.

4.7 Module 7 — Exam Supervision

4.7.1 Supervisor Assignment

- Admin Senior Executive assigns Exam Supervisors to specific exam slots (date, time, room).
- Supervisors are notified of their assignment via push notification.

4.7.2 Reporting Time Enforcement

- Supervisor must check in via the app before the exam start time.
- Late check-in is logged and flagged for payroll penalty computation.

4.7.3 Malpractice Reporting

- Supervisor can log a malpractice incident against a student: incident type, description, evidence upload.
- Incident is immediately visible to Admin Senior Executive and Branch Manager.

4.7.4 Hour-Based Compensation

- Each supervision slot has a defined duration (hours).
- Compensation is auto-computed based on hours supervised and linked to the Supervisor's payroll record.
- Late entry penalty is deducted per policy configuration.

4.8 Module 8 — Faculty & Payroll

4.8.1 Faculty Profile

- Profile fields: name, photo, qualifications, subject expertise, contact info, joined date, employment type (full-time / part-time / contract), bank details for payroll.

4.8.2 QR Attendance

- Faculty check-in and check-out using QR scan at institution premises (same infrastructure as student attendance).
- Late entry is logged and linked to payroll penalty computation.

4.8.3 Session Reporting

- After each teaching session, Faculty submits a session report via mobile app: subject, topic / chapter covered, completion percentage, remarks.
- Admin Senior Executive and Branch Manager can review submitted session reports.

4.8.4 Hour-Based Payroll Computation

- Payroll is computed based on hours taught (derived from QR attendance + session reports).
- Configurable: hourly rate per subject, late entry penalty deduction, absence deduction.

- Accountant runs payroll at month-end; Branch Manager approves before disbursement.
- Faculty receives a digital payroll slip via in-app notification.

4.9 Module 9 — Leave Management

4.9.1 Leave Types

Leave Type	Description
Paid Leave (PL)	Pre-approved leave deducted from the annual paid leave balance.
Sick Leave (SL)	Medical leave; may require a doctor's certificate for > 2 consecutive days.
Casual Leave (CL)	Short-notice or unforeseen absence.
Club / Special Leave	Institution-specific leave category (e.g., inter-college events, workshops).

4.9.2 Application & Approval Workflow

- Faculty submits leave request via mobile app: leave type, dates, reason, supporting document (optional).
- Multi-level approval chain: Admin Senior Executive → Branch Manager.
- System auto-notifies each approver in sequence via push notification.
- Applicant receives a push notification on approval or rejection with the reason.

4.9.3 Special Rules

- Half-Day Detection: If leave covers only a morning or afternoon session, system classifies it as a 0.5-day deduction.
- Sandwich Leave Policy: If leave is applied on days surrounding a weekend or public holiday, those intervening days are automatically counted as leave days.
- Late Entry Rules: Repeated late entries (configurable threshold — e.g., 3 late entries in a month) trigger an automatic half-day leave deduction.

4.10 Module 10 — In-App Chat

4.10.1 Chat Channels

Channel	Participants
Student ↔ Admin	Direct chat between a student and Admin/Branch Manager/Super Admin.
Parent ↔ Admin	Direct chat between a parent and Admin Senior Executive / Branch Manager / Super Admin.
Faculty — Group only	Faculty may participate in group/batch channels only; no one-to-one direct chat initiated by Faculty.

4.10.2 Feature Requirements

- Real-time message delivery (WebSocket or equivalent).
- Message history persisted and searchable.
- Read receipts: single tick (sent), double tick (delivered), filled tick (read).
- File sharing: images, PDFs, up to 10 MB per attachment.
- Push notification for every new unread message (see Module 11).
- Message timestamps displayed in the recipient's local timezone.

4.11 Module 11 — Notification System

All notifications are delivered via mobile push (FCM / APNs) and surfaced in an in-app Notification Centre. WhatsApp and SMS are out of scope.

4.11.1 Notification Events

#	Event	Recipient(s)	Channel
1	Student Check-In	Parent	Push + In-App
2	Student Check-Out	Parent	Push + In-App
3	Missing Scan (IN or OUT)	Parent	Push + In-App
4	15-Min Delay Alert	Parent	Push + In-App
5	30-Min Escalation	Parent + Admin	Push + In-App
6	Fee Due Reminder	Student + Parent	Push + In-App
7	Payment Confirmation	Student + Parent	Push + In-App
8	Digital Receipt	Student + Parent	Push + In-App
9	Exam Notification	Student	Push + In-App
10	Exam Results Published	Student + Parent	Push + In-App
11	Paper Checker Assignment	Checker	Email + In-App
12	Leave Approval / Rejection	Staff Member	Push + In-App
13	New Chat Message	Recipient	Push + In-App
14	Timetable Published	Student + Faculty	Push + In-App
15	Timetable Updated	Student + Faculty	Push + In-App
16	Cash Fee Approval (pending)	Branch Manager	Push + In-App
17	Cash Fee Approved	Admin Senior Executive	Push + In-App
18	Refund Approval Request	Branch Manager	Push + In-App
19	Payroll Slip Generated	Faculty / Supervisor / Checker	Push + In-App
20	Support Ticket Raised (Parent)	Super Admin / Branch Manager / Admin Sr Exec	Push + In-App

4.11.2 In-App Notification Centre

- Accessible from a bell icon in the app header.
- Displays full history of all notifications with timestamp and read/unread status.
- Tapping a notification navigates directly to the relevant screen/record.
- Mark-all-as-read action available.

4.12 Module 12 — Audit Logs

- Automatically captures all significant system events: fee approval / rejection, record creation / modification / deletion (with before-and-after field-level diff), leave approval / rejection.
- Each log entry records: event type, acting user, role, timestamp, affected record ID, before-state, after-state.
- Searchable admin viewer: filter by date range, event type, user, module, or record ID.
- Audit logs are read-only — no user can modify or delete a log entry.
- Available to Super Admin and Branch Manager only.

4.13 Module 13 — Reporting & Dashboard

4.13.1 Admin KPI Dashboard

- Total active students, new admissions (this month), dropout rate.
- Attendance rate (branch-wide, batch-wise).
- Fee collection summary: collected vs due, overdue amount.
- Upcoming exams and pending results.
- Open CRM leads by pipeline stage.

4.13.2 Reports

Report Name	Key Metrics
Fee Collection Report	Collected, pending, overdue amounts; payment-mode breakdown; student-wise drill-down.
Attendance Report	Daily / weekly / monthly attendance %; absentee list; delay / violation summary.
Timetable Utilisation Report	Classroom occupancy rates; faculty load; free-slot analysis.
Student Performance Report	Exam scores, percentile ranking, subject-wise analysis, recheck history.
Faculty Payroll Report	Hours taught, payroll computed, penalties, disbursement status.
CRM Conversion Report	Leads by stage, conversion rate, lost leads by reason, counsellor performance.
Leave Report	Leave balance per staff member, leave taken by type, pending approvals.

- All reports support date-range filters and batch/branch filters.
- Export options: PDF and CSV for all reports.

5. Key Business Rules Summary

Rule	Description
BR-001: Cash Fee Approval	Cash payment entries created by Admin Senior Executive must be approved by Branch Manager before they are posted to the student's account.
BR-002: Refund Mandatory Fields	No refund can be processed without a transaction ID (bank transfer) or cheque number. Branch Manager approval is mandatory.
BR-003: Sandwich Leave	If leave immediately precedes or follows a weekend or public holiday, all intervening non-working days are counted as leave days.
BR-004: Half-Day Leave	Leave applications covering only morning or afternoon sessions are counted as 0.5 days.
BR-005: Late Entry Penalty	Configurable threshold (default: 3 late entries per month) triggers an automatic 0.5-day leave deduction.
BR-006: Single Exam Attempt	Online exams allow only one attempt per student. Submission is final and cannot be retaken.
BR-007: Paper Recheck Flow	Student raises recheck → Admin Senior Executive reviews → assigned to Checker → result updated → student notified.
BR-008: Timetable Conflict Block	A timetable with unresolved conflicts cannot be published.
BR-009: Branch Data Isolation	Branch Manager and all roles below cannot access data from other branches.
BR-010: Audit Immutability	Audit log entries cannot be modified or deleted by any user, including Super Admin.
BR-011: Faculty Chat Restriction	Faculty cannot initiate or receive direct (one-to-one) chat; Faculty participates in group/batch channels only.
BR-012: QR Offline Queue	QR attendance scans collected while offline are queued locally and synced when connectivity is restored. Timestamps reflect the original scan time.

6. Document Control

Field	Value
Document Title	Functional Requirement Document — Education Management System Phase 1
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